

Raw SERS data



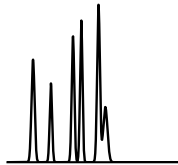
SERS map

one spectrum
per pixel



NNLS gate

background
dropped

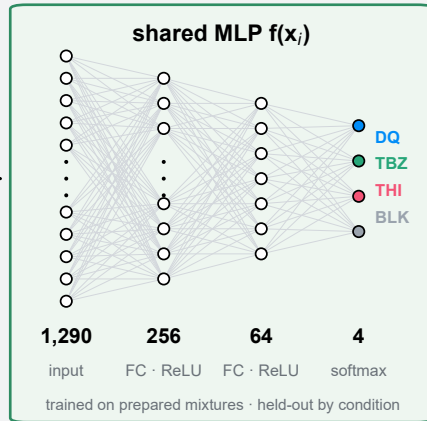


pixel spectra x_i

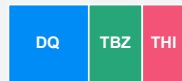
1,290 pts · 500–2500 cm^{-1}

Compositional unmixing

output



map composition



gated pixels · composition

$\bar{p}_i = \text{pixel mean} \cdot \text{BLK removed}$

absent component → reported 0

(absence gate, $P < 0.2$)

Concentration

output

reported concentration

calibration + correction net

$$\log_{10} \hat{C} = \log_{10} C_{cal} + \Delta$$

outside measured domain → no answer

declared total (optional)

$$\hat{C}_i = \bar{p}_i \cdot C_{total}$$

overlay when formulation known

domain guard: measured-map library (k-NN distance)